

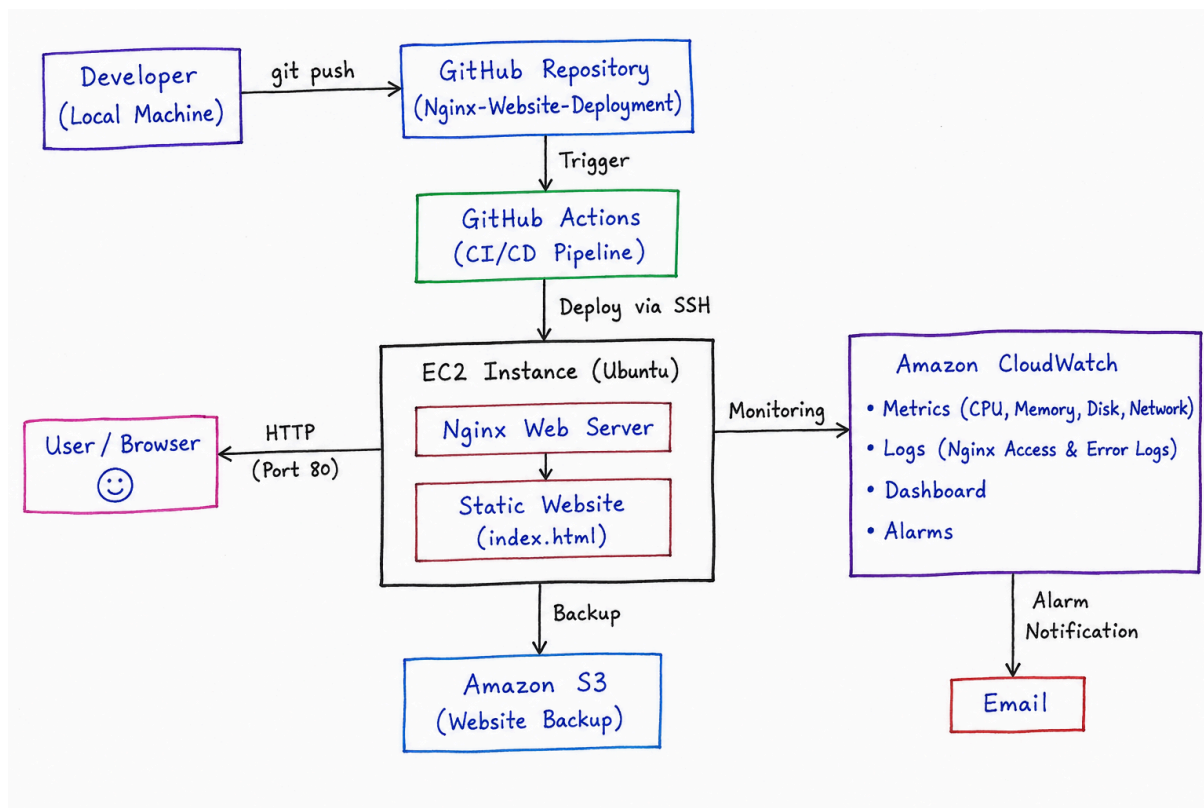
DevOps Engineer Technical Assignment

1. Project Overview

This deployment guide documents the complete deployment of a production-like web application on AWS using DevOps best practices.

The project includes the deployment of an Ubuntu EC2 instance running Nginx, an automated CI/CD pipeline using GitHub Actions, Amazon S3 backup, IAM role configuration, CloudWatch monitoring, dashboards, alarms, centralized log collection, and load testing using k6. The guide provides the implementation steps with supporting screenshots for each stage of the deployment.

2. Architecture Diagram



3. Amazon EC2 Instance Was Launched

Instance summary for i-0f78759c530fc4b8c (Ubuntu EC2) [Info](#)

Updated less than a minute ago

Instance ID

 i-0f78759c530fc4b8c

IPv6 address

—

Hostname type

IP name: ip-192-168-1-45.ap-south-1.compute.internal

Answer private resource DNS name

—

Auto-assigned IP address



IAM role

 DevOps-EC2-S3-Role [↗](#)

IMDSv2

Required

Public IPv4 address

 3.110.78.204 | [open address ↗](#)

Instance state

 **Running**

Private IP DNS name (IPv4 only)

 ip-192-168-1-45.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

 vpc-0c5511574a3fe2b6b (Personal-VPC) [↗](#)

Subnet ID

 subnet-0579133b5767b4e3d (Public subnet) [↗](#)

Instance ARN

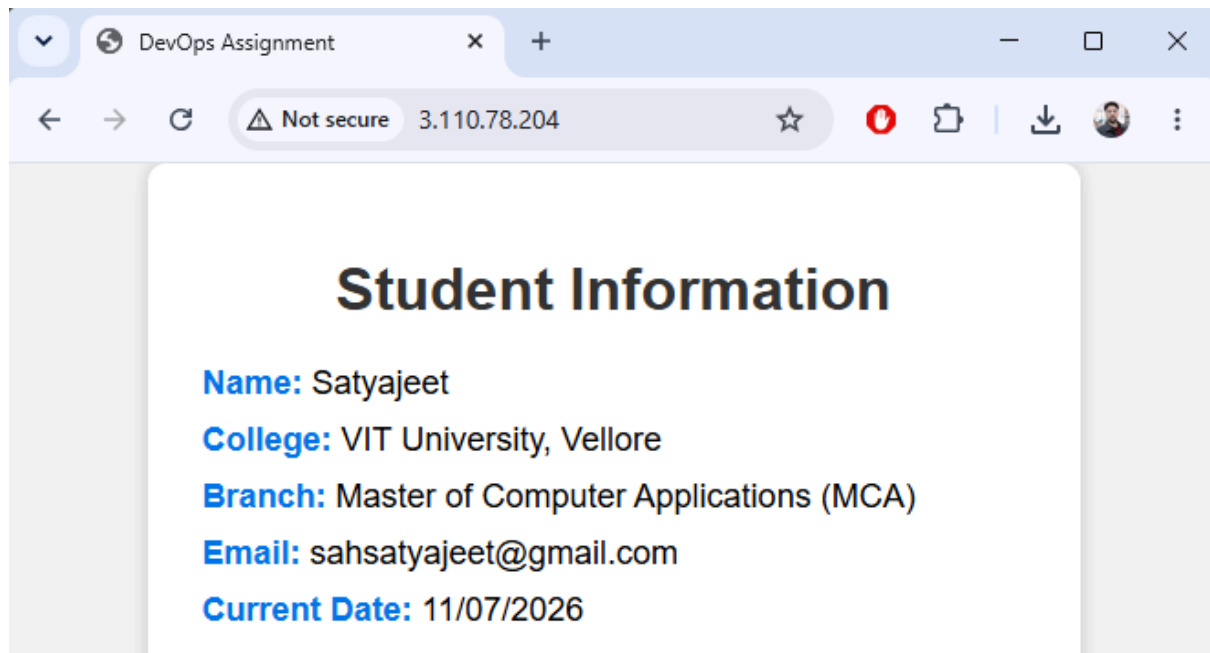
 arn:aws:ec2:ap-south-1:720294270515:instan

4. Nginx Was Installed

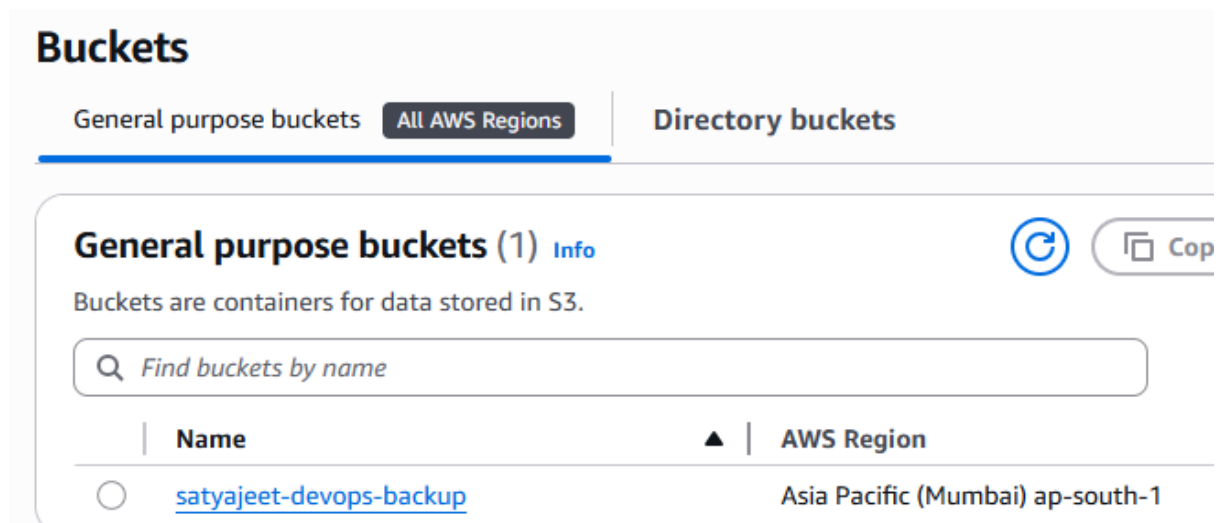
```
ubuntu@ip-192-168-1-45:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-10 21:58:14 UTC; 3 days ago
     Invocation: e70c6b22ebd948ccbe8388aa2add6de7
       Docs: man:nginx(8)
    Main PID: 41981 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 137.4M (peak: 139.5M)
        CPU: 2min 12.119s
    CGroup: /system.slice/nginx.service
            └─41981 "nginx: master process /usr/sbin/nginx -g daemon on; master_proce
               └─41982 "nginx: worker process"
                  └─41983 "nginx: worker process"

Jul 10 21:58:14 ip-192-168-1-45 systemd[1]: Starting nginx.service - A high performanc
Jul 10 21:58:14 ip-192-168-1-45 systemd[1]: Started nginx.service - A high performanc
ubuntu@ip-192-168-1-45:~$
```

5. Website Was Deployed Successfully



6. Amazon S3 Bucket Was Created



7. Website Backup Was Configured in Amazon S3

website-backup/

Objects

Properties

Objects (3)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3](#)

🔍 Find objects by prefix

☐	Name	▲	Type
☐	 index.html		html
☐	 index.nginx-debian.html		html

8. Website Backup Was Automated Using Cron (Runs every day at 2:00 AM)

```
ubuntu@ip-192-168-1-45:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /home/ubuntu/scripts/backup.sh >> /home/ubuntu/scripts/back
up.log 2>&1
ubuntu@ip-192-168-1-45:~$
```

9. IAM Policy Was Created

Modify permissions in S3BackupPolicy [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements

Policy editor

```
1  [
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "ListBucket",
6        "Effect": "Allow",
7        "Action": [
8          "s3:ListBucket",
9          "s3:GetBucketLocation"
10       ],
11       "Resource": "arn:aws:s3:::satyajeet-devops-backup"
12     },
13     {
14       "Sid": "ObjectAccess",
15       "Effect": "Allow",
16       "Action": [
17         "s3:GetObject",
18         "s3:PutObject",
19         "s3:DeleteObject"
20       ],
21       "Resource": "arn:aws:s3:::satyajeet-devops-backup/**"
22     }
23   ]
24 ]
```

10. IAM Role Was Created and Attached to the EC2 Instance

DevOps-EC2-S3-Role

Info

Allows EC2 instances to call AWS services on your behalf.

Summary

Creation date

July 12, 2026, 03:50 (UTC+05:30)

Last activity

✓

19 hours ago

ARN

arn:aws:iam::720294270

Maximum session duration

4 hours

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2)

Info

You can attach up to 10 managed policies.

Search

Policy name

▲

+

CloudWatchAgentServerPolicy

AWS managed

+

S3BackupPolicy

Customer managed

Details

Status and alarms

Monitoring

Security

▼ Security details

IAM role

DevOps-EC2-S3-Role

11. GitHub Repository Was Created

Find a repository...

Type

Language

Sort

New

Nginx-Website-Deployment

Public

☆

Star

Replaced the default Nginx welcome page with a custom HTML webpage displaying student information.

HTML

Updated 2 days ago

12. GitHub Actions CI/CD Pipeline Was Configured

All workflows
Showing runs from all workflows

 **Help us improve GitHub Actions**
Tell us how to make GitHub Actions work better for you with three quick questions.

4 workflow runs

 Updated homepage alignment to center using GitHub Actions Deploy Website to EC2 #4: Commit 10e7ba9 pushed by Satyajeeet-Sah	main
 Updated homepage alignment using GitHub Actions Deploy Website to EC2 #3: Commit d630bb8 pushed by Satyajeeet-Sah	main
 Updated homepage using GitHub Actions Deploy Website to EC2 #2: Commit da3ffbd pushed by Satyajeeet-Sah	main
 Added GitHub Actions CI/CD workflow via YAML file. Deploy Website to EC2 #1: Commit 901aed3 pushed by Satyajeeet-Sah	main

13. CloudWatch Agent Was Installed

```
ubuntu@ip-192-168-1-45:~$ sudo systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; vendor preset: enabled)
   Active: active (running) since Tue 2026-07-14 19:40:49 UTC; 1min ago
     Invocation: 1e0603634da7407fb312abde90045d34
       Main PID: 604 (amazon-cloudwat)
         Tasks: 8 (limit: 627)
        Memory: 139.1M (peak: 139.1M)
           CPU: 1.161s
        CGroup: /system.slice/amazon-cloudwatch-agent.service
               └─604 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent

Jul 14 19:40:51 ip-192-168-1-45 start-amazon-cloudwatch-agent[639]: >
```


14. CloudWatch Agent Configuration Was Created

```
] ,  
    "metrics_collection_interval": 10,  
    "resources": [  
        "*" ]  
},  
"mem": {  
    "measurement": [  
        "mem_used_percent"  
],  
    "metrics_collection_interval": 10  
},  
"swap": {  
    "measurement": [  
        "swap_used_percent"  
],  
    "metrics_collection_interval": 10 }  
  
}  
  
}
```

Please check the above content of the config.
The config file is also located at /opt/aws/amazon-cloudwatch-agent/bin/config.json.
Edit it manually if needed.

Do you want to store the config in the SSM parameter store?

1. yes

2. no

default choice: [1]:

2

Program exits now.

ubuntu@ip-192-168-1-45:~\$ █

15. CloudWatch Metrics Were Published Successfully

Browse (57)
Multi source query
Graphed metrics
Options
Source

All > CWAgent

☐ Alarm recommendations ⓘ

Download alarm code ▼
Create alarm
Graph with SQL
Graph search

Mumbai ▼

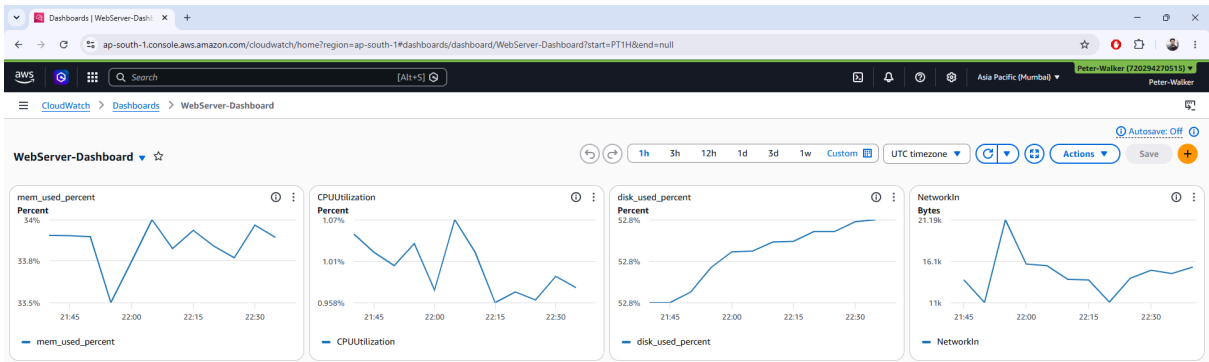
Search for any metric, dimension, resource id or account id

Imageld, InstanceId, InstanceType, device, fstype, path
44

Imageld, InstanceId, InstanceType, name
11

Imageld, InstanceId, InstanceType
2

16. CloudWatch Dashboard Was Created



17. CloudWatch Alarms Were Configured

Alarms (3)

Filter alarms

Quick filters

State = OK X Clear filters

☐	Name	State	Actions
☐	High-Memory-Alarm	OK	Actions enabled
☐	High-Disk-Usage-Alarm	OK	Actions enabled
☐	High_CPU_Alerts	OK	Actions enabled

18. Application Logs Were Collected in CloudWatch Logs

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

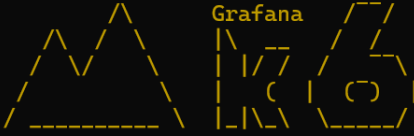
Filter events - press enter to search

Clear

	Timestamp	Message
		There are older events to load. Load more.
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:16.830Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:17.080Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"
▶	2026-07-13T23:19:17.080Z	49.47.132.15 - - [13/Jul/2026:23:19:16 +0000] "GET / HTTP/1.1" 200 1396 "-" "Grafana k6/2.0.0"

19. Load Testing Was Performed Using k6

```
satyajeet@corextech:~/k6-test$ k6 run load-test.js
```

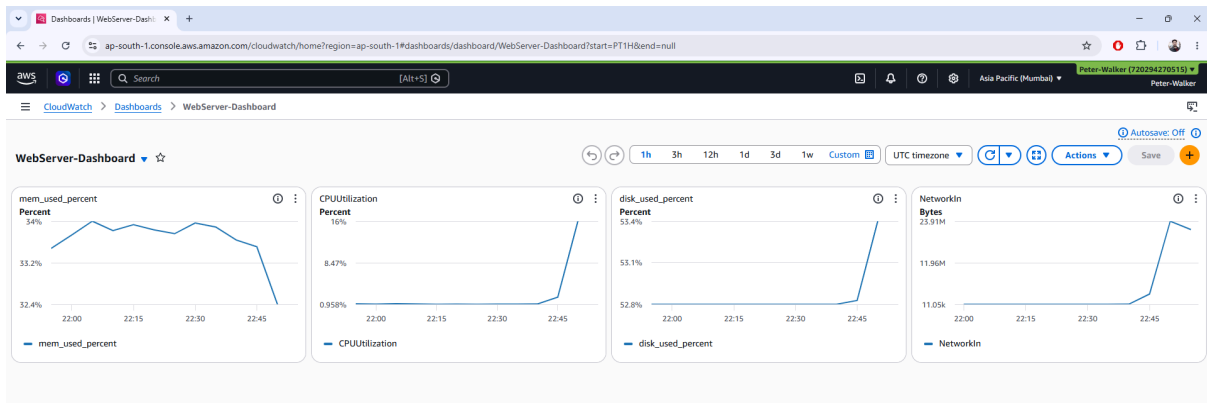


```
execution: local
script: load-test.js
output: -

scenarios: (100.00%) 1 scenario, 20 max VUs, 1m30s max duration (incl. graceful stop):
* default: 20 looping VUs for 1m0s (gracefulStop: 30s)

running (0m40.8s), 20/20 VUs, 700 complete and 0 interrupted iterations
default [=====>-----] 20 VUs 0m37.4s/1m0s
```

20. CloudWatch Dashboard Was Monitored During Load Testing



21. Performance Analysis Was Completed

```
satyajeet@corextech:~/k6-te  +  v
```

```
■ TOTAL RESULTS
```

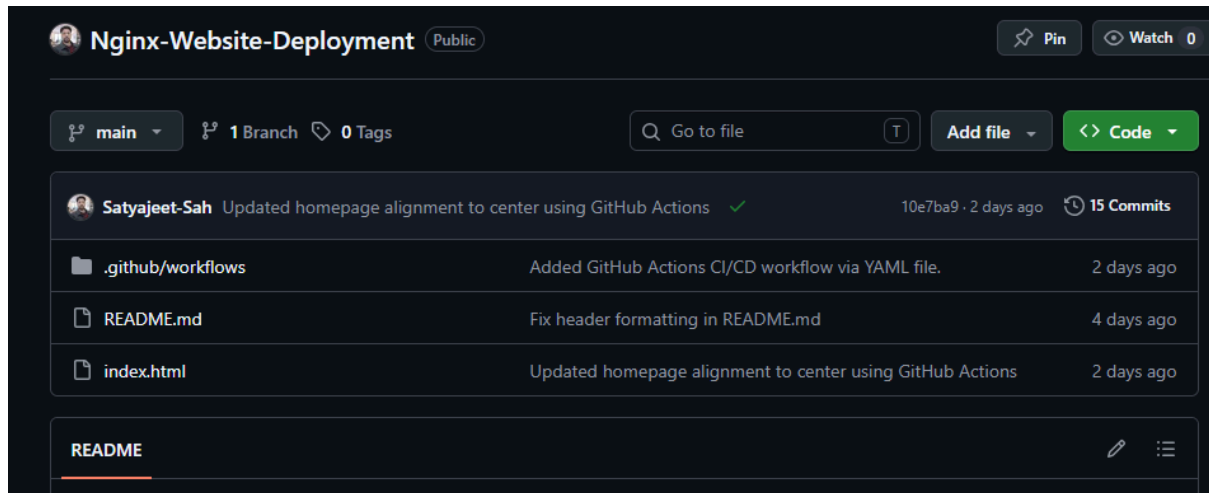
```
HTTP
http_req_duration.....: avg=50.5ms min=39.94ms med=47.04ms max=4.27s p(90)=52.48ms p(95)=59.65ms
{ expected_response:true }...: avg=50.5ms min=39.94ms med=47.04ms max=4.27s p(90)=52.48ms p(95)=59.65ms
http_req_failed.....: 0.00% 0 out of 171251
http_reqs.....: 171251 237.534674/s
```

```
EXECUTION
iteration_duration.....: avg=1.05s min=1.04s med=1.04s max=5.28s p(90)=1.05s p(95)=1.06s
iterations.....: 171251 237.534674/s
vus.....: 1 min=1 max=499
vus_max.....: 500 min=500 max=500
```

```
NETWORK
data_received.....: 282 MB 391 kB/s
data_sent.....: 12 MB 16 kB/s
```

```
running (12m01.0s), 000/500 VUs, 171251 complete and 0 interrupted iterations
default ✓ [=====] 000/500 VUs 12m0s
satyajeet@corextech:~/k6-test$
```

22. Final Project on GitHub



23. Conclusion

The project was successfully completed by deploying a production-like web application on AWS using DevOps best practices. The implementation included automated CI/CD using GitHub Actions, Amazon S3 backup, IAM role configuration, CloudWatch monitoring, dashboards, alarms, centralized log collection, and performance testing using k6. The deployed infrastructure was successfully monitored, tested, and documented.